

Retirement Income Planner - Technical Map

Executive Summary

The Retirement Income Planner is a Vue.js + Laravel module that calculates tax-optimised retirement income drawdown strategies. Users can interactively model their retirement income from multiple account types (pensions, ISAs, GIAs, savings) with real-time tax calculations, fund depletion projections, and tax-efficient allocation recommendations.

1. Account Types Used

The system gathers accounts from multiple sources and treats them differently for tax purposes:

Account Type	Source Model	Tax Treatment	Tax Rate	Notes
DC Pension - PCLS	DCPension	Tax-free	0%	25% of pot as lump sum
DC Pension - Drawdown	DCPension	Taxable	Marginal	Remaining 75%, taxed at marginal rate
DB Pension	DBPension	Taxable	Marginal	Fixed annual income, taxed at marginal rate
State Pension	StatePension	Taxable	Marginal	Only if retirement age >= 67
Onshore Bond	InvestmentAccount (onshore_bond)	Tax-deferred	5% tax-free	5% cumulative withdrawal rule; gains taxed on encashment
Offshore Bond	InvestmentAccount (offshore_bond)	Tax-deferred	5% tax-free	5% cumulative withdrawal rule; gross roll-up, taxed on encashment
Cash ISA	SavingsAccount (is_isa=true)	Tax-free	0%	No withdrawal restrictions
Stocks & Shares ISA	InvestmentAccount (isa type)	Tax-free	0%	No CGT or income tax
GIA	InvestmentAccount (gia type)	Taxable	Marginal	Subject to CGT and dividend tax
Savings (non-ISA)	SavingsAccount (is_isa=false)	Taxable	Marginal	Interest taxable, PSA may apply

Bond Account Tax Treatment

5% Cumulative Withdrawal Rule:

- Policyholders can withdraw up to 5% of the original investment each year without immediate tax
- Unused allowance rolls forward (e.g., if no withdrawals for 5 years, can withdraw 25% tax-free)
- Withdrawals within the 5% allowance are treated as "return of capital" not income
- Tax is deferred until full encashment or exceeding cumulative allowance

Onshore vs Offshore:

Feature	Onshore Bond	Offshore Bond
Fund growth	Taxed within fund at 20%	Gross roll-up (no internal tax)
Basic rate taxpayer	No further tax on encashment	20% tax on gains at encashment
Higher rate taxpayer	20% additional tax on gains	40% tax on gains at encashment
Top-slicing relief	Available	Available

"Marginal Rate": The exact tax rate depends on total income. The system uses **TaxBandTracker** to calculate based on how much room remains in each tax band.

GIA Account Tax Treatment

- GIA accounts will generally have a CGT tax liability
- We will not deduct any tax from a GIA account, however we will note it as taxable
- As the CGT is calculated elsewhere, and fluctuates, this account is lower on the priority of use list

2. Available Accounts Structure

When **getAvailableAccounts()** is called, it returns accounts with **projected values at retirement** using Monte Carlo 80% confidence (20th percentile). All drawable assets are projected forward to the user's target retirement age.

Projection Methods by Account Type

Account Type	Projection Method	Source
Pension Pot (combined DC)	Monte Carlo 80%	<code>RetirementProjectionService::projectPensionPot()</code> → <code>percentile_20_at_retirement</code>

Account Type	Projection Method	Source
Stocks & Shares ISA	Monte Carlo 80%	InvestmentProjectionService::getAccountProjectedValue80()
GIA	Monte Carlo 80%	InvestmentProjectionService::getAccountProjectedValue80()
Onshore Bond	Monte Carlo 80%	InvestmentProjectionService::getAccountProjectedValue80()
Offshore Bond	Monte Carlo 80%	InvestmentProjectionService::getAccountProjectedValue80()
Cash ISA	Compound growth	2% annual growth rate
DB Pension	N/A	Income-based (no pot value)
State Pension	N/A	Income-based (no pot value)

Pension Pot (Combined DC Pensions with sub-accounts)

All DC pensions are combined into a single "Pension Pot" using the Monte Carlo 80% projected value. This represents the combined value of ALL DC pensions at retirement age.

```
[
  {
    'id' => 'pension_pot',
    'type' => 'pension_pot',
    'owner_id' => $userId,
    'name' => 'Pension Pot',
    'value' => 485000.00,           // Monte Carlo 80% projected value at retirement
    'pcls_available' => 121250.00, // 25% of projected value (tax-free)
    'tax_treatment' => 'taxable',
    'is_projected' => true,
    'sub_accounts' => [
      {
        'source_type' => 'pension_pot_pcls',
        'source_id' => 'pension_pot',
        'name' => 'Pension Pot - Tax-Free Cash (PCLS)',
        'max_amount' => 121250.00, // 25% of projected pot
        'tax_rate' => 0,
        'tax_treatment' => 'tax_free'
      },
      {
        'source_type' => 'pension_pot_drawdown',
        'source_id' => 'pension_pot',
        'name' => 'Pension Pot - Drawdown',
        'max_amount' => 363750.00, // 75% of projected pot
        'tax_rate' => null,         // Marginal rate
        'tax_treatment' => 'taxable'
      }
    ]
  }
]
```

DB Pension

DB pensions are income-based (no pot value to project).

```
[
  {
    'id' => 1,
    'type' => 'db_pension',
    'owner_id' => $userId,
    'name' => 'NHS Pension',
    'provider' => 'NHS',
    'value' => null,           // No pot value
    'annual_income' => 15000.00, // Fixed annual income
    'payment_start_age' => 60,
    'lump_sum_entitlement' => 45000.00,
    'tax_treatment' => 'taxable',
    'source_type' => 'db_pension',
    'source_id' => 1
  }
]
```

State Pension

State pensions are income-based (no pot value to project).

```
[
  {
    'id' => 1,
    'type' => 'state_pension',
  }
]
```

```

'owner_id' => $userId,
'name' => 'State Pension',
'value' => null,
'annual_income' => 11502.00,           // Full new state pension
'payment_start_age' => 67,
'already_receiving' => false,
'tax_treatment' => 'taxable',
'source_type' => 'state_pension',
'source_id' => 1
]

```

ISA - Stocks & Shares (Monte Carlo 80%)

```

[
  'id' => 1,
  'type' => 'isa_investment',
  'owner_id' => $userId,
  'name' => 'Vanguard ISA',
  'platform' => 'Vanguard',
  'current_value' => 100000.00,       // Current value today
  'value' => 156420.00,               // Monte Carlo 80% projected value
  'is_projected' => true,
  'years_projected' => 15,
  'projection_type' => 'monte_carlo_80',
  'isa_type' => 'stocks_shares',
  'tax_rate' => 0,
  'tax_treatment' => 'tax_free',
  'source_type' => 'isa',
  'source_id' => 1
]

```

ISA - Cash (Compound Growth)

```

[
  'id' => 2,
  'type' => 'isa_cash',
  'owner_id' => $userId,
  'name' => 'Marcus Cash ISA',
  'current_value' => 20000.00,       // Current value today
  'value' => 26930.00,               // Projected at 2% growth
  'is_projected' => true,
  'years_projected' => 15,
  'growth_rate' => 0.02,              // 2% for cash
  'isa_type' => 'cash',
  'tax_rate' => 0,
  'tax_treatment' => 'tax_free',
  'source_type' => 'isa',
  'source_id' => 2
]

```

GIA (Monte Carlo 80%)

```

[
  'id' => 1,
  'type' => 'gia',
  'owner_id' => $userId,
  'name' => 'Hargreaves Lansdown GIA',
  'platform' => 'Hargreaves Lansdown',
  'current_value' => 50000.00,       // Current value today
  'value' => 78210.00,               // Monte Carlo 80% projected value
  'is_projected' => true,
  'years_projected' => 15,
  'projection_type' => 'monte_carlo_80',
  'tax_rate' => null,                 // Marginal rate (CGT applies)
  'tax_treatment' => 'taxable',
  'source_type' => 'gia',
  'source_id' => 1
]

```

Onshore Bond (Monte Carlo 80%)

```
[
  'id' => 1,
  'type' => 'onshore_bond',
  'owner_id' => $userId,
  'name' => 'Prudential Onshore Bond',
  'provider' => 'Prudential',
  'current_value' => 100000.00,      // Current value today
  'value' => 156420.00,             // Monte Carlo 80% projected value
  'is_projected' => true,
  'years_projected' => 15,
  'projection_type' => 'monte_carlo_80',
  'original_investment' => 80000.00, // For 5% calculation
  'annual_tax_free_allowance' => 4000.00, // 5% of original investment
  'tax_rate' => 0,                  // Within 5% allowance
  'tax_treatment' => 'tax_deferred',
  'source_type' => 'onshore_bond',
  'source_id' => 1
]
```

Offshore Bond (Monte Carlo 80%)

```
[
  'id' => 1,
  'type' => 'offshore_bond',
  'owner_id' => $userId,
  'name' => 'RL360 Offshore Bond',
  'provider' => 'RL360',
  'current_value' => 150000.00,      // Current value today
  'value' => 234630.00,             // Monte Carlo 80% projected value
  'is_projected' => true,
  'years_projected' => 15,
  'projection_type' => 'monte_carlo_80',
  'original_investment' => 100000.00,
  'annual_tax_free_allowance' => 5000.00, // 5% of original investment
  'tax_rate' => 0,                  // Within 5% allowance
  'tax_treatment' => 'tax_deferred',
  'source_type' => 'offshore_bond',
  'source_id' => 1
]
```

Key Fields for Projected Accounts

Field	Description
current_value	The account's value today
value	The projected value at retirement (used for calculations)
is_projected	Boolean flag indicating value is projected
years_projected	Number of years until retirement
projection_type	Either <code>monte_carlo_80</code> or compound growth rate

3. Tax Optimisation Strategy

3.1 Default Allocation Algorithm

When the user loads the planner, `calculateDefaultAllocations()` runs the following:

```
Check if user has any Guaranteed Income (can't avoid) from
├─ State Pension (if retirement_age >= 67)
├─ DB Pension (if retirement_age >= normal_retirement_age)
  These use Personal Allowance first, reducing overall tax, and work out any potential tax user may have

This will reduce the target income needed (either user entered or default) by the after tax income received from
above, giving us a net target income to work with.
We will use this 'net target income' as the starting point for the fund depletion algorithm, which is run at this
point.
Once this is run, and we have confirmed either the net target income or a reduced income due to insufficient funds,
we can start to draw from the various funds as follows:

Tax-Free/Tax-Deferred Sources (Bonds, PCLS, ISA)
├─ Bonds: 5% of original investment (tax-deferred, uses cumulative allowance)
├─ Fill remaining personal allowance(if state pension and any db pension payments left any personal allowance) with
```

```

drawdown
├ PCLS (100% tax-free)
├ ISA accounts, first cash ISA then stocks & shares ISA(100% tax-free)
├ Savings withdrawals (treat as return of capital)
  No immediate tax impact

Taxable Flexible Income
├ DC Drawdown (additional)
├ GIA withdrawals

  Fills remaining target at basic/higher/additional rates

```

If any accounts or items are missing, go to the next step, so if the user has not entered any state pension details, and has no DB pension, run the depletion algorithm, and carry on from that point.

3.2 Tax Efficiency Priority Order

We must work out the optimal tax efficient balance for the user between PCLS and ISA withdrawal to maximise their tax free draw for as long as possible.

1. Always use full personal allowance each year
If the state pension and/or db payments do not use the full personal allowance, ensure that any remaining personal allowance is used up from drawdown.
Do not go above the tax years personal allowance limit, for 2025/26 this is £12,570
2. State Pension / DB Pension
├ Guaranteed, unavoidable – uses PA first
3. Bond Withdrawals (5% tax-deferred)
├ Uses cumulative 5% allowance – no immediate tax
├ Preserves tax-deferred growth in remaining bond value
4. ISA Withdrawals (cash isa before stocks & shares)
├ 100% tax-free, preserves tax-advantaged growth
├ After bonds as ISA has no cumulative limit
5. DC Pension Drawdown
├ Use remaining PA, then basic rate
6. GIA / Savings
├ Last resort – highest marginal rates

3.3 Tax Band Application

The **TaxBandTracker** service applies income to tax bands in order:

Personal Allowance: £12,570 (2025/26) └ First £12,570 of taxable income = 0% tax
Basic Rate: £12,571 – £50,270 └ Next £37,700 = 20% tax
Higher Rate: £50,271 – £125,140 └ Next £74,870 = 40% tax
Additional Rate: Over £125,140 └ Everything above = 45% tax

Note: Personal Allowance tapers for income over £100,000 (reduced by £1 for every £2 over £100k).

4. Fund Depletion Projection

Algorithm

Need to run multiple scenarios on the users income need (target or net target income) and the funds they have available (pension funds by default). This is done everytime a user includes or excludes an asset from the 'Other Assets' section. This must also run everytime the monte carlo projections changes for any of the included pensions and/or assets.

projectFundDepletion() calculates year-by-year fund balances from retirement to age 100:

```

For each year from retirement_age to 100:
  1. Deduct annual withdrawals from each fund for the included assets
    - 5% bond withdrawal

```

- DC drawdown up to any remaining personal allowance
- DC Pension: Deduct PCLS
- ISA: Deduct Cash ISA then stocks & shares ISA withdrawal
- GIA: Deduct GIA withdrawal
- Savings: Deduct savings withdrawal
- Drawdown: taxable portion over and above the personal allowance

2. Check for depletion

- If balance <= 0, record depletion age
- If the age is < 95 and > 105, run the algorithm again with different values to get within this range

3. Apply growth (if balance > 0)

- Pension: 4% growth
- ISA: 4% growth
- GIA: 4% growth
- Cash/Savings: 0% growth

4. Aggregate by fund type for chart

- dc_pension: Sum all DC pension balances
- isa: Sum all Cash ISA balances
- isa: Sum all Stocks & Shares ISA balances
- gia: Sum all GIA balances
- savings: Sum all savings balances

5. Investment Account Retirement Inclusion

Before an investment account appears in the Retirement Income Planner Sources of Income section, it must be explicitly marked for inclusion. This prevents all investment accounts from automatically appearing in retirement calculations.

Account Filtering Logic

```
getAvailableAccounts():
```

1. Query InvestmentAccounts WHERE user_id = \$userId

2. Filter: WHERE include_in_retirement = true

3. Only accounts with this flag enabled are:

- Shown in the Retirement Income Planner sources of income section
- Included in fund depletion projections

4. Accounts with this flag disabled are shown in the 'Other Assets' section.

Database Field

Field	Type	Default	Description
include_in_retirement	boolean	false	Whether account is included in retirement planning

User Control

Users can toggle this setting per investment account via a toggle shown on the card in the retirement planner other assets section.

Affected Account Types

This filter applies to:

- Stocks & Shares ISA
- Cash ISA (via SavingsAccount)
- GIA (General Investment Account)
- Onshore Bonds
- Offshore Bonds
- LISA (Lifetime ISA)

Rationale

Not all investment accounts are suitable for retirement funding:

- Short-term savings goals (e.g., house deposit in ISA)
- Emergency funds in accessible accounts
- Accounts earmarked for other purposes (education, gifts)
- Illiquid investments with lock-in periods

By requiring explicit opt-in, users maintain control over which assets form part of their retirement income strategy.

6. Rules

Rule 1: Use PMT Formula to Deplete Tax-Free accounts at Age 100

Tax-free accounts should reach £0 at age 100. Use PMT formula for growing accounts(bonds being an exception for now):

PMT Formula: $\text{Annual Withdrawal} = \text{Balance} \times (r \times (1+r)^n) / ((1+r)^n - 1)$

Where:

- r = growth rate (4% for investments, 0% for cash)
- n = years to age 100

PCLS Annual = PCLS Total ÷ Years to 100 (no growth – simple division)

Bond Annual = PMT(bond_balance, 0.04, years_to_100)

ISA Annual = PMT(isa_balance, growth_rate, years_to_100)

- Investment ISA: 4% growth
- Cash ISA: 0% growth (simple division)

Rule 2: PMT Multipliers by Years

Quick reference for 4% growth:

Years to 100	PMT Multiplier
35	0.0536
38	0.0508
40	0.0505
42	0.0495
45	0.0483

Example: £200,000 ISA, 35 years to 100
PMT = £200,000 × 0.0536 = £10,720/year

Rule 3: Only Reduce Income If Funds Deplete Before 100

1. Simulate depletion with target(or net target) income using PMT allocations
2. Check: Do funds hit £0 BEFORE age 100?

IF funds > £0 at age 100:

- Use target(or net target) income
- Tax-free accounts deplete to £0
- Pension/GIA may have remaining balance (acceptable)

IF funds < £0 BEFORE age 95:

- Income exceeds sustainable level
- Show warning: "Income will drop at age X"
- Reduce target(or net target) income to sustainable level

7. Year by year table

The table must reflect the following for each year from retirement to 100:

- year, showing the age of the user
- withdrawal amount - adjusted for any tax, so if the user's target income is £10k, and there is £1k tax, the withdrawal will be £11k
- columns for the various accounts included, if an account is excluded this column must also disappear, and the table calculation redone and if included the column will appear and calculations redone
- pcls use
- drawdown use
- tax payable
- total balance of all included accounts

8. Testing Protocol

This section documents the comprehensive testing procedures for verifying the Retirement Income Planner functionality.

8.1 Prerequisites

1. **Database Seeded:** Run `php artisan db:seed` to ensure all preview personas have data
2. **Dev Server Running:** Start with `./dev.sh` or `php artisan serve + npm run dev`
3. **Test Persona:** Use "David & Sarah Mitchell" (peak_earners) - they have comprehensive pension and investment data

8.2 Access Path

1. Navigate to `http://localhost:8000`
2. Click "Try the Demo"

- 3. Select "David & Sarah Mitchell" persona
- 4. From Dashboard, click "Pensions" in the Assets section
- 5. Click "Retirement" in the left sidebar
- 6. Click on the "Retirement Income Planner" card

8.3 Core Value Verification

Verify the following values are displayed correctly:

Field	Expected Behaviour
Pension Capital at Retirement	Shows Monte Carlo 80% projected value (e.g., £901,562)
PCLS	25% of Pension Capital (e.g., £225,390)
Pension Drawdown	75% of Pension Capital (e.g., £676,171)
Other Assets at Retirement	Sum of included ISA/GIA/Savings projected values
Optimised Income	Adjusted if funds deplete before age 100, otherwise equals target
Annual Tax	£0 while drawing from tax-free sources

8.4 Account Toggle Testing

The "Other Assets" section contains toggles to include/exclude accounts. Test each combination:

8.4.1 Single Account Tests

Test	Toggle ON	Expected Changes
ISA Only	David's S&S ISA	Other Assets increases, Income Sources shows ISA card, Gap to Target reduces
GIA Only	Joint GIA	Other Assets increases, Income Sources shows GIA card, GIA column appears in table
Savings Only	HSBC/Nationwide	Other Assets increases, Savings column appears in chart/table
Cash ISA Only	Nationwide Cash ISA	Other Assets increases, ISA column shows combined value

8.4.2 Combination Tests

Test	Accounts Included	Verify
All Tax-Free	ISA + Cash ISA	Tax remains £0, ISA column shows combined, depletion order correct
Tax-Free + Taxable	ISA + GIA	Tax appears when taxable sources drawn, correct depletion order
All Accounts	ISA + GIA + All Savings	Maximum income achieved, all columns visible
Pension Only	None toggled on	Only PCLS + Drawdown shown, Other Assets = £0

8.4.3 Toggle Verification Checklist

For EACH toggle change, verify:

- ☐ **Summary Cards Update:** Optimised Income, Other Assets, Gap to Target recalculate
- ☐ **Income Sources Update:** New account card appears/disappears with projected value and annual draw
- ☐ **Chart Updates:** Legend shows/hides account type, stacked areas adjust
- ☐ **Table Updates:** Column appears/disappears for toggled account type
- ☐ **Tax Recalculates:** Tax changes appropriately based on taxable/tax-free mix

8.5 Fund Depletion Chart Verification

The stacked area chart should display:

Element	Verification
Legend	Shows all included account types (PCLS, Pension Drawdown, ISA, GIA, Savings)
X-axis	Age from retirement (e.g., 60) to 100
Y-axis	Fund Value in £k
State Pension Age	Vertical annotation at age 67
Depletion Order	Visual confirmation: PCLS depletes first, then ISA, then GIA/Savings, Drawdown last
End Balance	Funds reach near £0 at age 100 (or show remaining balance)

8.6 Year-by-Year Table Verification

The depletion table must show:

Column	Content
Age	Sequential from retirement age to 100
Withdrawal	Annual withdrawal amount (red, negative)
PCLS	Balance with withdrawal shown below (depletes first)
Drawdown	Balance with withdrawal shown below
ISA	Balance with withdrawal shown below (if included)
GIA	Balance with withdrawal shown below (if included)
Savings	Balance with withdrawal shown below (if included)
Growth	Annual growth amount (green, positive)
Taxable Drawdown	Amount exceeding Personal Allowance
Tax Paid	Tax amount (red when > 0, green £0 when tax-free)
Total Balance	Sum of all fund balances

8.6.1 Depletion Order Verification

Verify in the table that accounts deplete in this order:

1. **PCLS depletes first** - Should reach £0 within first few years (e.g., by age 64-65)
2. **ISA depletes second** - After PCLS exhausted, ISA draws begin (e.g., depletes by age 72)
3. **GIA/Savings deplete third** - If included, after ISA exhausted
4. **Pension Drawdown last** - Continues until age 100

8.6.2 Tax Transition Verification

- **Tax-free phase:** While drawing from PCLS + ISA, Tax Paid should be £0
- **Tax transition:** When taxable drawdown exceeds PA, tax appears (e.g., age 72+)
- **Consistent tax:** Once in taxable phase, tax should be consistent each year (e.g., £8,704/year)

8.7 Edge Case Tests

Test Case	Setup	Expected
No State Pension	User has no state pension entered	Warning shown, projections exclude state pension
Already Retired	Retirement age <= current age	Uses current values, not projected
Insufficient Funds	Small pension, high target income	Income adjusted down, warning shown
Very High Income	Target £150k+	Additional rate tax calculated
Spouse Included	Toggle "Include Spouse"	Combined assets shown, joint calculations

8.8 Calculation Accuracy Tests

8.8.1 PMT Formula Verification

For a given ISA balance and years to retirement:

$$\text{PMT} = \text{Balance} \times (r \times (1+r)^n) / ((1+r)^n - 1)$$

Where $r = 0.04$ (4%), $n = \text{years to age 100}$

Example verification:

- ISA Balance at retirement: £339,510
- Years to age 100: 40
- Expected PMT ≈ £17,153/year (verify Annual draw shown matches)

8.8.2 Tax Calculation Verification

When taxable drawdown occurs:

$$\text{Taxable amount} = \text{Drawdown} - \text{remaining Personal Allowance}$$

$$\text{Tax} = \text{Taxable amount} \times 20\% \text{ (basic rate)}$$

Example:

- Total drawdown: £56,092

- State Pension: £11,502 (uses PA first)
- Remaining PA: £12,570 - £11,502 = £1,068
- Taxable: £56,092 - £1,068 = ~£55,024
- Tax at 20%: ~£11,005

8.9 Regression Test Checklist

After any code changes, verify:

- ☐ Pension pot uses 80% Monte Carlo projection
- ☐ PCLS is exactly 25% of pension pot
- ☐ Drawdown is exactly 75% of pension pot
- ☐ ISA toggle adds account to Income Sources
- ☐ GIA toggle adds account to Income Sources
- ☐ Chart legend updates when accounts toggled
- ☐ Table columns appear/disappear correctly
- ☐ Tax is £0 during tax-free withdrawal phase
- ☐ Tax appears when taxable withdrawals begin
- ☐ Funds project to age 100 (not depleting early unless income adjusted)
- ☐ Warning message appears if income was adjusted

8.10 Browser Console Checks

Open browser DevTools (F12) and check:

1. **No JavaScript errors** in Console tab
2. **API responses successful** in Network tab:
 - `GET /api/retirement/income` returns 200
 - `PATCH /api/investments/{id}` returns 200 when toggling
3. **Correct data in responses:**
 - `projected_pension_pot` is populated
 - `available_accounts` includes pension_pot with sub_accounts
 - `fund_projections` array has entries from retirement age to 100

Scenarios

SCENARIO 1: Simple ISA + Pension

Profile:

- Age: 65, Retirement Age: 65, Life Expectancy: 100 (35 years)
- Target Income: £30,000/year

Assets:

Asset	Balance	Tax Treatment
DC Pension	£400,000	PCLS: Tax-free, Drawdown: Taxable
S&S ISA	£200,000	Tax-free

Calculations:

- PCLS Total: £400,000 × 25% = £100,000
- PCLS Annual (deplete at 100): £100,000 ÷ 35 = £2,857/year
- Pension Drawdown Pool: £400,000 × 75% = £300,000
- ISA PMT (4% growth, 35 years): £200,000 × 0.0536 = £10,720/year

Tax-Free Total: £2,857 + £10,720 = £13,577/year

Remaining from taxable: £30,000 - £13,577 = £16,423/year

Annual Allocation:

Source	Amount	Tax Treatment	Tax
PCLS	£2,857	Tax-free	£0
ISA	£10,720	Tax-free	£0
Pension Drawdown	£16,423	Taxable	£770.60
Total	£30,000		£770.60

Tax Calculation:

- Tax-free income: £2,857 + £10,720 = £13,577
- Taxable income: £16,423
- Personal Allowance used: £12,570 (£0 tax)
- Basic Rate: £16,423 - £12,570 = £3,853 × 20% = £770.60

Year-by-Year Projection (Years 1-10, then key years):

Year	Age	PCLS W/D	ISA W/D	Pension W/D	Total W/D	PCLS Bal	ISA Bal	Pension Bal	Total Funds
0	65	-	-	-	-	£100,000	£200,000	£300,000	£600,000
1	66	£2,857	£10,720	£16,423	£30,000	£97,143	£196,851	£294,920	£588,914
2	67	£2,857	£10,720	£16,423	£30,000	£94,286	£193,605	£289,713	£577,604
3	68	£2,857	£10,720	£16,423	£30,000	£91,429	£190,259	£284,378	£566,066
4	69	£2,857	£10,720	£16,423	£30,000	£88,572	£186,810	£278,910	£554,292
5	70	£2,857	£10,720	£16,423	£30,000	£85,715	£183,252	£273,303	£542,270
10	75	£2,857	£10,720	£16,423	£30,000	£71,430	£164,031	£244,031	£479,492
15	80	£2,857	£10,720	£16,423	£30,000	£57,145	£141,631	£211,254	£410,030
20	85	£2,857	£10,720	£16,423	£30,000	£42,860	£115,515	£174,481	£332,856
25	90	£2,857	£10,720	£16,423	£30,000	£28,575	£85,013	£133,139	£246,727
30	95	£2,857	£10,720	£16,423	£30,000	£14,290	£49,383	£86,554	£150,227
35	100	£2,857	£10,720	£16,423	£30,000	£0	£0	£33,906	£33,906

Math Check (Year 1):

- ISA: (£200,000 - £10,720) × 1.04 = £196,851 ✓
- Pension: (£300,000 - £16,423) × 1.04 = £294,920 ✓
- PCLS: £100,000 - £2,857 = £97,143 (no growth) ✓

Result: PCLS and ISA deplete to £0 at age 100. Pension has £33,906 remaining (buffer). Tax reduced from £1,035 to £770.60 by using more ISA.

SCENARIO 2: Bond-Heavy Portfolio

Profile:

- Age: 60, Retirement Age: 60, Life Expectancy: 100 (40 years)
- Target Income: £40,000/year

Assets:

Asset	Balance	Original Investment	Tax Treatment
Onshore Bond	£300,000	£200,000	5% tax-deferred
Offshore Bond	£250,000	£180,000	5% tax-deferred
DC Pension	£350,000	-	PCLS/Drawdown
Cash ISA	£50,000	-	Tax-free

Calculations:

- Onshore Bond PMT (4%, 40 yrs): £300,000 × 0.0505 = £15,150/year
- Offshore Bond PMT (4%, 40 yrs): £250,000 × 0.0505 = £12,625/year
- PCLS Total: £350,000 × 25% = £87,500
- PCLS Annual: £87,500 ÷ 40 = £2,188/year
- Pension Drawdown Pool: £350,000 × 75% = £262,500
- Cash ISA (no growth): £50,000 ÷ 40 = £1,250/year

Tax-Free/Deferred Total: £15,150 + £12,625 + £2,188 + £1,250 = £31,213/year

Note: Bond 5% limits are:

- Onshore: £200,000 × 5% = £10,000 (withdrawing £15,150 exceeds this)
- Offshore: £180,000 × 5% = £9,000 (withdrawing £12,625 exceeds this)

The excess over 5% triggers a "chargeable event" but is still more tax-efficient than pension drawdown for basic rate taxpayers.

Remaining from taxable: £40,000 - £31,213 = £8,787/year

Annual Allocation:

Source	Amount	Tax Treatment	Tax
Onshore Bond	£15,150	Tax-deferred*	£0
Offshore Bond	£12,625	Tax-deferred*	£0
PCLS	£2,188	Tax-free	£0
Cash ISA	£1,250	Tax-free	£0
Pension Drawdown	£8,787	Taxable	£0
Total	£40,000		£0

*Bond withdrawals within policy, tax deferred until full encashment

Tax Calculation:

- Taxable: £8,787
- Personal Allowance: £12,570 covers all
- Total Tax: £0

Year-by-Year Projection:

Year	Age	Onshore W/D	Offshore W/D	PCLS W/D	ISA W/D	Pension W/D	Total	Onshore Bal	Offshore Bal	PCLS Bal	ISA Bal	Pension Bal	Total
0	60	-	-	-	-	-	-	£300,000	£250,000	£87,500	£50,000	£262,500	£950,000
1	61	£15,150	£12,625	£2,188	£1,250	£8,787	£40,000	£296,244	£246,870	£85,312	£48,750	£263,862	£941,038
5	65	£15,150	£12,625	£2,188	£1,250	£8,787	£40,000	£280,907	£234,089	£76,560	£43,750	£268,579	£903,885
10	70	£15,150	£12,625	£2,188	£1,250	£8,787	£40,000	£257,618	£214,682	£65,620	£37,500	£276,171	£851,591
20	80	£15,150	£12,625	£2,188	£1,250	£8,787	£40,000	£195,031	£162,526	£43,740	£25,000	£299,632	£725,929
30	90	£15,150	£12,625	£2,188	£1,250	£8,787	£40,000	£108,512	£90,427	£21,860	£12,500	£335,416	£568,715
40	100	£15,150	£12,625	£2,188	£1,250	£8,787	£40,000	£0	£0	£0	£0	£388,125	£388,125

Result: ZERO TAX! All tax-free accounts deplete at 100. Pension grows to £388k as reserve. Excellent tax efficiency.

SCENARIO 3: High Income Requirement

Profile:

- Age: 55, Retirement Age: 55, Life Expectancy: 100 (45 years)
- Target Income: £80,000/year

Assets:

Asset	Balance	Tax Treatment
DC Pension	£800,000	PCLS/Drawdown
S&S ISA	£300,000	Tax-free
GIA	£150,000	Taxable (CGT)

Step 1: Calculate Tax-Free Withdrawals (to deplete at 100)

- PCLS Total: £800,000 × 25% = £200,000
- PCLS Annual: £200,000 ÷ 45 = £4,444/year
- ISA PMT (4%, 45 yrs): £300,000 × 0.0483 = £14,490/year

Tax-Free Total: £4,444 + £14,490 = £18,934/year

Step 2: Calculate Taxable Withdrawal Needed

Remaining from taxable: £80,000 - £18,934 = £61,066/year

Step 3: Check Sustainability of Taxable Sources

- Pension Drawdown Pool: £600,000
- GIA: £150,000
- Total Taxable Pool: £750,000

Sustainable PMT from £750,000 at 4% over 45 years: PMT = £750,000 × 0.0483 = £36,225/year

Problem: We need £61,066/year but can only sustain £36,225/year!

This means taxable accounts WILL deplete before age 100.

Step 4: Calculate When Taxable Accounts Deplete

Combined taxable (£750,000) at 4% growth, withdrawing £61,066/year:

Using annuity formula: $n \approx 17$ years

Taxable accounts deplete around age 72.

Step 5: Split Taxable Withdrawal Between Pension and GIA

To deplete BOTH taxable sources together at age 72 (17 years):

- Pension PMT (4%, 17 yrs): $£600,000 \times 0.0822 = £49,320/\text{year}$
- GIA PMT (4%, 17 yrs): $£150,000 \times 0.0822 = £12,330/\text{year}$
- Total: $£61,650/\text{year} \approx £61,066$ needed ✓

Corrected Annual Allocation (Ages 55-71):

Source	Amount	Tax Treatment	Tax
PCLS	£4,444	Tax-free	£0
ISA	£14,490	Tax-free	£0
Pension Drawdown	£48,853	Taxable	Higher Rate
GIA	£12,213	Taxable	Higher Rate
Total	£80,000		£11,858

Tax Calculation:

- Tax-free: $£4,444 + £14,490 = £18,934$
- Taxable: $£48,853 + £12,213 = £61,066$
- Personal Allowance: $£12,570 @ 0\%$
- Basic Rate: $£37,700 @ 20\% = £7,540$
- Higher Rate: $£61,066 - £50,270 = £10,796 @ 40\% = £4,318.40$
- **Total Tax: £11,858.40**

Year-by-Year Projection (Phase 1: Ages 55-71):

Year	Age	PCLS	ISA	Pension	GIA	Total	PCLS Bal	ISA Bal	Pension Bal	GIA Bal	Total Funds
0	55	-	-	-	-	-	£200,000	£300,000	£600,000	£150,000	£1,250,000
1	56	£4,444	£14,490	£48,853	£12,213	£80,000	£195,556	£296,930	£573,193	£143,298	£1,208,977
5	60	£4,444	£14,490	£48,853	£12,213	£80,000	£177,780	£282,787	£487,656	£121,914	£1,070,137
10	65	£4,444	£14,490	£48,853	£12,213	£80,000	£155,560	£262,994	£366,510	£91,628	£876,692
15	70	£4,444	£14,490	£48,853	£12,213	£80,000	£133,340	£239,362	£219,036	£54,759	£646,497
17	72	£4,444	£14,490	£48,853	£12,213	£80,000	£124,452	£227,001	£130,892	£32,723	£515,068

Phase 2: After Taxable Sources Deplete (Ages 72-100)

Around age 72-73, pension and GIA both reach £0. Only tax-free sources remain:

- PCLS remaining: ~£115,000
- ISA remaining: ~£215,000

New Annual Income (Tax-Free Only):

Source	Amount	Tax
PCLS	£4,444	£0
ISA	£14,490	£0
Total	£18,934	£0

Income Drops from £80,000 to £18,934!

Full Projection (Ages 72-100):

Year	Age	PCLS	ISA	Total	PCLS Bal	ISA Bal	Total Funds
18	73	£4,444	£14,490	£18,934	£120,008	£221,091	£341,099
25	80	£4,444	£14,490	£18,934	£88,900	£186,238	£275,138
35	90	£4,444	£14,490	£18,934	£44,460	£127,350	£171,810

Year	Age	PCLS	ISA	Total	PCLS Bal	ISA Bal	Total Funds
45	100	£4,444	£14,490	£18,934	£0	£0	£0

Summary:

Phase	Ages	Annual Income	Annual Tax	Notes
Phase 1	55-72	£80,000	£11,858	All sources used
Phase 2	73-100	£18,934	£0	Tax-free only

Key Insight: £80,000/year is NOT sustainable to age 100 with these assets. The user has two choices:

- 1. **Accept income drop:** £80k for 17 years, then £19k for 28 years
- 2. **Reduce target:** Sustainable income to age 100 = ~£55,000/year

Sustainable Income Calculation:

If target was reduced to deplete ALL accounts at 100:

- PCLS: £4,444/year
- ISA PMT: £14,490/year
- Pension PMT: £600,000 × 0.0483 = £28,980/year
- GIA PMT: £150,000 × 0.0483 = £7,245/year
- **Total Sustainable: £55,159/year**

At £55,159/year:

- Tax-free: £18,934
- Taxable: £36,225
- Tax: £36,225 - £12,570 = £23,655 × 20% = £4,731/year (all basic rate!)

Result: High income (£80k) cannot be sustained. Either accept income cliff at 72, or reduce to sustainable £55k.

SCENARIO 4: Zero Tax Achievable

Profile:

- Age: 60, Retirement Age: 60, Life Expectancy: 100 (40 years)
- Target Income: £25,000/year

Assets:

Asset	Balance	Original Investment	Tax Treatment
DC Pension	£300,000	-	PCLS/Drawdown
S&S ISA	£400,000	-	Tax-free
Onshore Bond	£200,000	£150,000	5% tax-deferred

Calculations:

- Bond PMT (4%, 40 yrs): £200,000 × 0.0505 = £10,100/year
- PCLS Total: £300,000 × 25% = £75,000
- PCLS Annual: £75,000 ÷ 40 = £1,875/year
- ISA PMT (4%, 40 yrs): £400,000 × 0.0505 = £20,200/year
- Pension Drawdown Pool: £225,000

Tax-Free Total: £10,100 + £1,875 + £20,200 = £32,175/year

Since £32,175 > £25,000, we can achieve **ZERO TAX** and not touch pension!

Adjusted Allocation (to hit exactly £25,000):

Reduce ISA withdrawal: £25,000 - £10,100 - £1,875 = £13,025/year from ISA

Source	Amount	Tax Treatment	Tax
Bond	£10,100	Tax-deferred	£0
PCLS	£1,875	Tax-free	£0
ISA	£13,025	Tax-free	£0
Pension Drawdown	£0	-	£0
Total	£25,000		£0

Year-by-Year Projection:

Year	Age	Bond W/D	PCLS W/D	ISA W/D	Pension W/D	Total	Bond Bal	PCLS Bal	ISA Bal	Pension Bal	Total
0	60	-	-	-	-	-	£200,000	£75,000	£400,000	£225,000	£900,000
1	61	£10,100	£1,875	£13,025	£0	£25,000	£197,496	£73,125	£402,454	£234,000	£907,075
5	65	£10,100	£1,875	£13,025	£0	£25,000	£186,881	£65,625	£410,960	£273,914	£937,380
10	70	£10,100	£1,875	£13,025	£0	£25,000	£170,594	£56,250	£422,406	£333,171	£982,421
20	80	£10,100	£1,875	£13,025	£0	£25,000	£129,050	£37,500	£451,123	£493,095	£1,110,768
30	90	£10,100	£1,875	£13,025	£0	£25,000	£71,877	£18,750	£491,247	£730,145	£1,312,019
40	100	£10,100	£1,875	£13,025	£0	£25,000	£0	£0	£546,621	£1,080,985	£1,627,606

Wait - ISA is growing, not depleting!

The ISA withdrawal (£13,025) is less than growth (£400,000 × 4% = £16,000), so ISA grows.

Recalculate to deplete ISA at 100:

Since we need to deplete tax-free accounts, increase ISA withdrawal and keep pension at £0.

New ISA withdrawal needed to reach £0 at 100: $PMT = £400,000 \times 0.0505 = £20,200/\text{year}$

But total would be: £10,100 + £1,875 + £20,200 = £32,175 > £25,000 target

Solution: Reduce bond withdrawal to balance:

- Target: £25,000
- PCLS: £1,875 (fixed to deplete)
- ISA: £20,200 (PMT to deplete)
- Bond: £25,000 - £1,875 - £20,200 = £2,925/year

Corrected Allocation:

Source	Amount	Tax Treatment	Tax
Bond	£2,925	Tax-deferred	£0
PCLS	£1,875	Tax-free	£0
ISA	£20,200	Tax-free	£0
Pension Drawdown	£0	-	£0
Total	£25,000		£0

Corrected Year-by-Year:

Year	Age	Bond W/D	PCLS W/D	ISA W/D	Pension W/D	Total	Bond Bal	PCLS Bal	ISA Bal	Pension Bal	Total
0	60	-	-	-	-	-	£200,000	£75,000	£400,000	£225,000	£900,000
1	61	£2,925	£1,875	£20,200	£0	£25,000	£204,958	£73,125	£395,392	£234,000	£907,475
10	70	£2,925	£1,875	£20,200	£0	£25,000	£232,866	£56,250	£343,809	£333,171	£966,096
20	80	£2,925	£1,875	£20,200	£0	£25,000	£290,049	£37,500	£261,296	£493,095	£1,081,940
30	90	£2,925	£1,875	£20,200	£0	£25,000	£385,422	£18,750	£147,856	£730,145	£1,282,173
40	100	£2,925	£1,875	£20,200	£0	£25,000	£546,098	£0	£0	£1,080,985	£1,627,083

Problem: Bond is GROWING because we're only taking £2,925 when growth is £8,000+/year.

Final Solution: We must accept that with £25k target and these assets, we cannot deplete all accounts. The optimal strategy is:

1. Deplete PCLS first (no growth): £1,875/year, depletes at 100 ✓
2. Deplete ISA via PMT: £20,200/year, depletes at 100 ✓
3. Remaining £2,925 from bond OR pension

Choose bond (tax-deferred) over pension (taxable). Bond will not deplete but that's acceptable.

Result: Zero tax achieved. PCLS and ISA deplete at 100. Bond and Pension are reserves. Funds grow to £1.6M+.

SCENARIO 5: Multiple Bonds Portfolio

Profile:

- Age: 58, Retirement Age: 58, Life Expectancy: 100 (42 years)
- Target Income: £50,000/year

Assets:

Asset	Balance	Original Investment	Tax Treatment
Prudential Onshore	£180,000	£120,000	Tax-deferred
Aviva Onshore	£150,000	£100,000	Tax-deferred
RL360 Offshore	£220,000	£160,000	Tax-deferred
DC Pension	£450,000	-	PCLS/Drawdown
S&S ISA	£100,000	-	Tax-free

Calculations (PMT to deplete at 100):

- Prudential PMT (4%, 42 yrs): £180,000 × 0.0495 = £8,910/year
- Aviva PMT (4%, 42 yrs): £150,000 × 0.0495 = £7,425/year
- RL360 PMT (4%, 42 yrs): £220,000 × 0.0495 = £10,890/year
- PCLS Total: £450,000 × 25% = £112,500
- PCLS Annual: £112,500 ÷ 42 = £2,679/year
- ISA PMT (4%, 42 yrs): £100,000 × 0.0495 = £4,950/year
- Pension Drawdown Pool: £337,500

Tax-Free/Deferred Total: £8,910 + £7,425 + £10,890 + £2,679 + £4,950 = £34,854/year

Remaining from taxable: £50,000 - £34,854 = £15,146/year

Annual Allocation:

Source	Amount	Tax Treatment	Tax
Prudential Bond	£8,910	Tax-deferred	£0
Aviva Bond	£7,425	Tax-deferred	£0
RL360 Bond	£10,890	Tax-deferred	£0
PCLS	£2,679	Tax-free	£0
ISA	£4,950	Tax-free	£0
Pension Drawdown	£15,146	Taxable	£515.20
Total	£50,000		£515.20

Tax Calculation:

- Taxable: £15,146
- Personal Allowance: £12,570 @ 0%
- Basic Rate: £15,146 - £12,570 = £2,576 × 20% = £515.20

Year-by-Year Projection:

Year	Age	Bonds Total	PCLS	ISA	Pension	Total	Prudential	Aviva	RL360	PCLS Bal	ISA Bal	Pension Bal	Total
0	58	-	-	-	-	-	£180,000	£150,000	£220,000	£112,500	£100,000	£337,500	£1,100,000
1	59	£27,225	£2,679	£4,950	£15,146	£50,000	£177,934	£148,278	£217,474	£109,821	£98,892	£335,248	£1,087,647
10	68	£27,225	£2,679	£4,950	£15,146	£50,000	£159,654	£133,045	£195,133	£85,710	£87,040	£319,988	£980,570
20	78	£27,225	£2,679	£4,950	£15,146	£50,000	£125,855	£104,879	£153,822	£59,010	£68,512	£294,759	£806,837
30	88	£27,225	£2,679	£4,950	£15,146	£50,000	£75,041	£62,534	£91,717	£32,310	£42,103	£257,780	£561,485
42	100	£27,225	£2,679	£4,950	£15,146	£50,000	£0	£0	£0	£0	£0	£203,821	£203,821

Result: Only £515.20 tax on £50k income (1.03% effective rate). All tax-free accounts deplete at 100. Pension remains as buffer.

SCENARIO 6: Large ISA Portfolio

Profile:

- Age: 62, Retirement Age: 62, Life Expectancy: 100 (38 years)
- Target Income: £35,000/year

Assets:

Asset	Balance	Tax Treatment
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Asset	Balance	Tax Treatment
S&S ISA (Vanguard)	£500,000	Tax-free
Cash ISA	£80,000	Tax-free
DC Pension	£200,000	PCLS/Drawdown

Calculations (PMT to deplete at 100):

- PCLS Total: £200,000 × 25% = £50,000
- PCLS Annual: £50,000 ÷ 38 = £1,316/year
- S&S ISA PMT (4%, 38 yrs): £500,000 × 0.0508 = £25,400/year
- Cash ISA (no growth): £80,000 ÷ 38 = £2,105/year
- Pension Drawdown Pool: £150,000

Tax-Free Total: £1,316 + £25,400 + £2,105 = £28,821/year

Remaining from taxable: £35,000 - £28,821 = £6,179/year

Annual Allocation:

Source	Amount	Tax Treatment	Tax
PCLS	£1,316	Tax-free	£0
S&S ISA	£25,400	Tax-free	£0
Cash ISA	£2,105	Tax-free	£0
Pension Drawdown	£6,179	Taxable	£0
Total	£35,000		£0

Tax Calculation:

- Taxable: £6,179
- Personal Allowance: £12,570 covers all
- Total Tax: £0

Year-by-Year Projection:

Year	Age	PCLS	S&S ISA	Cash ISA	Pension	Total	PCLS Bal	S&S ISA Bal	Cash ISA Bal	Pension Bal	Total
0	62	-	-	-	-	-	£50,000	£500,000	£80,000	£150,000	£780,000
1	63	£1,316	£25,400	£2,105	£6,179	£35,000	£48,684	£493,584	£77,895	£149,574	£769,737
10	72	£1,316	£25,400	£2,105	£6,179	£35,000	£36,840	£430,115	£58,950	£145,034	£670,939
20	82	£1,316	£25,400	£2,105	£6,179	£35,000	£23,680	£336,790	£37,900	£136,975	£535,345
30	92	£1,316	£25,400	£2,105	£6,179	£35,000	£10,520	£205,088	£16,850	£124,313	£356,771
38	100	£1,316	£25,400	£2,105	£6,179	£35,000	£0	£0	£0	£107,185	£107,185

Result: ZERO TAX achieved. All tax-free accounts deplete at 100. Pension remains as emergency buffer.

SCENARIO 7: Pension-Dominant Portfolio (DB + State Pension)

Profile:

- Age: 67, Retirement Age: 67, Life Expectancy: 100 (33 years)
- Target Income: £55,000/year
- State Pension: £11,502/year (immediate)
- DB Pension: £18,000/year (NHS pension, immediate)

Assets:

Asset	Balance/Income	Tax Treatment
DC Pension (SIPP)	£600,000	PCLS/Drawdown
State Pension	£11,502/year	Taxable (guaranteed)
DB Pension (NHS)	£18,000/year	Taxable (guaranteed)
S&S ISA	£50,000	Tax-free

Guaranteed Income: £11,502 + £18,000 = £29,502/year (taxable, unavoidable)

Calculations:

- PCLS Total: £600,000 × 25% = £150,000
- PCLS Annual: £150,000 ÷ 33 = £4,545/year
- ISA PMT (4%, 33 yrs): £50,000 × 0.0540 = £2,700/year
- Pension Drawdown Pool: £450,000

Tax-Free Available: £4,545 + £2,700 = £7,245/year

Remaining needed: £55,000 - £29,502 - £7,245 = £18,253/year (from DC Drawdown)

Annual Allocation:

Source	Amount	Tax Treatment	Tax
State Pension	£11,502	Taxable	Uses PA
DB Pension (NHS)	£18,000	Taxable	Basic Rate
PCLS	£4,545	Tax-free	£0
ISA	£2,700	Tax-free	£0
DC Pension Drawdown	£18,253	Taxable	Basic Rate
Total	£55,000		£6,937

Tax Calculation:

- Tax-free: £4,545 + £2,700 = £7,245
- Taxable: £11,502 + £18,000 + £18,253 = £47,755
- Personal Allowance: £12,570 @ 0%
- Basic Rate: £47,755 - £12,570 = £35,185 × 20% = £7,037

Year-by-Year Projection:

Year	Age	State P	DB Pension	PCLS	ISA	DC Drawdown	Total	PCLS Bal	ISA Bal	DC Bal	Total Funds
0	67	-	-	-	-	-	-	£150,000	£50,000	£450,000	£650,000
1	68	£11,502	£18,000	£4,545	£2,700	£18,253	£55,000	£145,455	£49,192	£448,617	£643,264
10	77	£11,502	£18,000	£4,545	£2,700	£18,253	£55,000	£104,550	£40,717	£433,880	£579,147
20	87	£11,502	£18,000	£4,545	£2,700	£18,253	£55,000	£59,100	£28,277	£408,648	£496,025
33	100	£11,502	£18,000	£4,545	£2,700	£18,253	£55,000	£0	£0	£366,104	£366,104

Result: Tax unavoidable due to guaranteed DB/State income using Personal Allowance. PCLS and ISA deplete at 100. DC remains as buffer.

SCENARIO 8: Early Retirement with State Pension Gap

Profile:

- Age: 55, Retirement Age: 55, State Pension Age: 67 (12-year gap)
- Life Expectancy: 100 (45 years)
- Target Income: £45,000/year

Assets:

Asset	Balance	Tax Treatment
DC Pension (SIPP)	£500,000	PCLS/Drawdown
S&S ISA	£250,000	Tax-free
GIA	£100,000	Taxable
State Pension (from 67)	£11,502/year	Taxable

Phase 1 (Ages 55-66): No State Pension - 12 years

- PCLS Total: £500,000 × 25% = £125,000
- PCLS Annual: £125,000 ÷ 45 = £2,778/year
- ISA PMT (4%, 45 yrs): £250,000 × 0.0483 = £12,075/year
- Pension Drawdown Pool: £375,000

Tax-Free: £2,778 + £12,075 = £14,853/year

Remaining: £45,000 - £14,853 = £30,147/year (from pension drawdown)

Phase 1 Allocation:

Source	Amount	Tax Treatment	Tax
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Source	Amount	Tax Treatment	Tax
PCLS	£2,778	Tax-free	£0
ISA	£12,075	Tax-free	£0
Pension Drawdown	£30,147	Taxable	£3,515.40
Total	£45,000		£3,515.40

Tax (Phase 1):

- Taxable: £30,147
- PA: £12,570 @ 0%
- Basic: £30,147 - £12,570 = £17,577 × 20% = £3,515.40

Phase 2 (Ages 67-100): State Pension Active - 33 years

State Pension: £11,502/year

Phase 2 Allocation:

Source	Amount	Tax Treatment	Tax
State Pension	£11,502	Taxable	Uses PA
PCLS	£2,778	Tax-free	£0
ISA	£12,075	Tax-free	£0
Pension Drawdown	£18,645	Taxable	£3,515.40
Total	£45,000		£3,515.40

Tax (Phase 2):

- Taxable: £11,502 + £18,645 = £30,147
- PA: £12,570 @ 0%
- Basic: £17,577 × 20% = £3,515.40

Year-by-Year Projection:

Year	Age	State P	PCLS	ISA	Pension	Total	PCLS Bal	ISA Bal	Pension Bal	GIA Bal	Total
0	55	£0	-	-	-	-	£125,000	£250,000	£375,000	£100,000	£850,000
1	56	£0	£2,778	£12,075	£30,147	£45,000	£122,222	£247,282	£358,647	£104,000	£832,151
5	60	£0	£2,778	£12,075	£30,147	£45,000	£111,110	£235,795	£299,580	£121,665	£768,150
12	67	£11,502	£2,778	£12,075	£18,645	£45,000	£91,664	£211,396	£207,243	£163,194	£673,497
20	75	£11,502	£2,778	£12,075	£18,645	£45,000	£69,440	£180,195	£177,982	£218,470	£646,087
30	85	£11,502	£2,778	£12,075	£18,645	£45,000	£41,660	£134,377	£137,668	£323,252	£636,957
45	100	£11,502	£2,778	£12,075	£18,645	£45,000	£0	£0	£60,234	£619,788	£680,022

Result: Same tax in both phases. PCLS and ISA deplete at 100. Pension and GIA remain as reserves.

SCENARIO 9: ISA-Only Early Years (Preserve Pension)

Profile:

- Age: 55, Retirement Age: 55, Life Expectancy: 100 (45 years)
- Target Income: £30,000/year
- Strategy: Maximise tax-free ISA withdrawals, let pension compound

Assets:

Asset	Balance	Tax Treatment
DC Pension	£400,000	PCLS/Drawdown
S&S ISA	£350,000	Tax-free

Strategy Analysis:

Option A: Use ISA + Pension together (standard approach) Option B: Use ISA only first, then pension (preserve pension growth)

Option B Calculation:

ISA-only years until ISA depletes:

- Target: £30,000/year from ISA
- ISA with 4% growth, £30,000 withdrawal

PMT formula reversed: How many years until £350,000 depletes at £30,000/year with 4% growth?

Using annuity formula: $n = \ln(1 - PV \times r / PMT) / \ln(1 + r)$... approximately 15 years

ISA depletes around age 70.

Meanwhile pension grows: $£400,000 \times 1.04^{15} = £720,366$

Year-by-Year (ISA-Only Phase, Years 1-15):

Year	Age	ISA W/D	Pension W/D	Total	ISA Bal	Pension Bal	Total Funds
0	55	-	-	-	£350,000	£400,000	£750,000
1	56	£30,000	£0	£30,000	£332,800	£416,000	£748,800
5	60	£30,000	£0	£30,000	£263,040	£486,661	£749,701
10	65	£30,000	£0	£30,000	£166,579	£592,098	£758,677
14	69	£30,000	£0	£30,000	£66,682	£688,493	£755,175
15	70	£30,000	£0	£30,000	£39,349	£716,033	£755,382
16	71	£30,000	£0	£30,000	£10,923	£744,674	£755,597
17	72	£10,923	£19,077	£30,000	£0	£754,541	£754,541

ISA depletes at age 72.

Post-ISA Phase (Age 72-100): Pension Only

Pension at age 72: £754,541

- PCLS: $£754,541 \times 25\% = £188,635$
- PCLS Annual: $£188,635 \div 28 = £6,737/\text{year}$
- Drawdown Pool: $£754,541 \times 75\% = £565,906$

Post-ISA Allocation:

Source	Amount	Tax Treatment	Tax
PCLS	£6,737	Tax-free	£0
Pension Drawdown	£23,263	Taxable	£2,138.60
Total	£30,000		£2,138.60

Tax (Post-ISA):

- Taxable: £23,263
- PA: £12,570 @ 0%
- Basic: $£10,693 \times 20\% = £2,138.60$

Comparison:

Strategy	Tax Years 1-17	Tax Years 18-45	Total Tax (45 yrs)
ISA-Only First	£0	$£2,138.60/\text{yr} \times 28 = £59,881$	£59,881
Standard Blend	$\sim £770/\text{yr} \times 45 = £34,650$	Same	£34,650

Result: ISA-only strategy pays MORE total tax! The standard blended approach is more tax-efficient because it spreads taxable income across more years, using Personal Allowance each year.

SCENARIO 10: Complex Multi-Asset Portfolio

Profile:

- Age: 60, Retirement Age: 60, State Pension Age: 67
- Life Expectancy: 100 (40 years)
- Target Income: £60,000/year

Assets:

Asset	Balance	Original Inv.	Tax Treatment
DC Pension (Workplace)	£350,000	-	PCLS/Drawdown

Asset	Balance	Original Inv.	Tax Treatment
DC Pension (SIPP)	£280,000	-	PCLS/Drawdown
S&S ISA	£180,000	-	Tax-free
Cash ISA	£45,000	-	Tax-free
Onshore Bond	£120,000	£90,000	Tax-deferred
Offshore Bond	£85,000	£60,000	Tax-deferred
GIA	£75,000	-	Taxable
Savings	£30,000	-	Taxable
State Pension (from 67)	£11,502/year	-	Taxable

Calculations (PMT to deplete at 100):

Pensions:

- Total DC Pension: £350,000 + £280,000 = £630,000
- PCLS Total: £630,000 × 25% = £157,500
- PCLS Annual: £157,500 ÷ 40 = £3,938/year
- Pension Drawdown Pool: £472,500

Bonds:

- Onshore PMT (4%, 40 yrs): £120,000 × 0.0505 = £6,060/year
- Offshore PMT (4%, 40 yrs): £85,000 × 0.0505 = £4,293/year

ISAs:

- S&S ISA PMT (4%, 40 yrs): £180,000 × 0.0505 = £9,090/year
- Cash ISA (no growth): £45,000 ÷ 40 = £1,125/year

Tax-Free/Deferred Total: £3,938 + £6,060 + £4,293 + £9,090 + £1,125 = £24,506/year

Phase 1 (Ages 60-66): No State Pension

Remaining from taxable: £60,000 - £24,506 = £35,494/year

Phase 1 Allocation:

Source	Amount	Tax Treatment	Tax
Onshore Bond	£6,060	Tax-deferred	£0
Offshore Bond	£4,293	Tax-deferred	£0
PCLS	£3,938	Tax-free	£0
S&S ISA	£9,090	Tax-free	£0
Cash ISA	£1,125	Tax-free	£0
Pension Drawdown	£35,494	Taxable	£4,584.80
Total	£60,000		£4,584.80

Tax (Phase 1):

- Taxable: £35,494
- PA: £12,570 @ 0%
- Basic: £35,494 - £12,570 = £22,924 × 20% = £4,584.80

Phase 2 (Ages 67-100): State Pension Active

State Pension: £11,502/year (uses PA first)

Phase 2 Allocation:

Source	Amount	Tax Treatment	Tax
State Pension	£11,502	Taxable	Uses PA
Onshore Bond	£6,060	Tax-deferred	£0
Offshore Bond	£4,293	Tax-deferred	£0
PCLS	£3,938	Tax-free	£0
S&S ISA	£9,090	Tax-free	£0

Source	Amount	Tax Treatment	Tax
Cash ISA	£1,125	Tax-free	£0
Pension Drawdown	£23,992	Taxable	£4,584.80
Total	£60,000		£4,584.80

Tax (Phase 2):

- Taxable: £11,502 + £23,992 = £35,494
- PA: £12,570 @ 0%
- Basic: £22,924 × 20% = £4,584.80

Year-by-Year Projection:

Yr	Age	State	Bonds	PCLS	ISAs	Pension	Total	Onshore	Offshore	PCLS Bal	S&S ISA	Cash ISA	Pension Bal	GIA
0	60	£0	-	-	-	-	-	£120,000	£85,000	£157,500	£180,000	£45,000	£472,500	£75,000
1	61	£0	£10,353	£3,938	£10,215	£35,494	£60,000	£118,498	£83,935	£153,562	£177,587	£43,875	£454,486	£78,000
7	67	£11,502	£10,353	£3,938	£10,215	£23,992	£60,000	£109,231	£77,369	£130,234	£161,788	£37,125	£431,584	£98,358
10	70	£11,502	£10,353	£3,938	£10,215	£23,992	£60,000	£103,416	£73,250	£118,358	£152,612	£33,750	£424,523	£110,611
20	80	£11,502	£10,353	£3,938	£10,215	£23,992	£60,000	£78,251	£55,428	£79,108	£115,502	£22,500	£392,883	£163,866
30	90	£11,502	£10,353	£3,938	£10,215	£23,992	£60,000	£43,562	£30,856	£39,858	£64,310	£11,250	£349,115	£242,647
40	100	£11,502	£10,353	£3,938	£10,215	£23,992	£60,000	£0	£0	£0	£0	£0	£289,612	£359,148

Result: All tax-free accounts (Bonds, PCLS, ISAs) deplete to £0 at age 100. Pension, GIA, and Savings remain as reserves. Same tax in both phases due to same total taxable income.

Summary Comparison

Scenario	Target	Assets	Tax/Year	Effective Rate	Tax-Free Depleted?
1. Simple ISA + Pension	£30,000	£600,000	£771	2.6%	Yes at 100
2. Bond-Heavy	£40,000	£950,000	£0	0%	Yes at 100
3. High Income	£80,000	£1,250,000	£11,858→£0	14.8%→0%	Taxable at 72, tax-free at 100
4. Zero Tax	£25,000	£900,000	£0	0%	PCLS+ISA at 100
5. Multiple Bonds	£50,000	£1,100,000	£515	1.0%	Yes at 100
6. Large ISA	£35,000	£780,000	£0	0%	Yes at 100
7. Pension-Dominant	£55,000	£650,000 + DB/SP	£7,037	12.8%	Yes at 100
8. Early Retirement	£45,000	£850,000	£3,515	7.8%	Yes at 100
9. ISA-Only Strategy	£30,000	£750,000	£0→£2,139	0→7.1%	ISA early, PCLS late
10. Complex Multi-Asset	£60,000	£1,165,000	£4,585	7.6%	Yes at 100